Diplomado en Inteligencia Artificial

Pontificia Universidad Católica de Chile

Tópicos de profundización

Razonamiento

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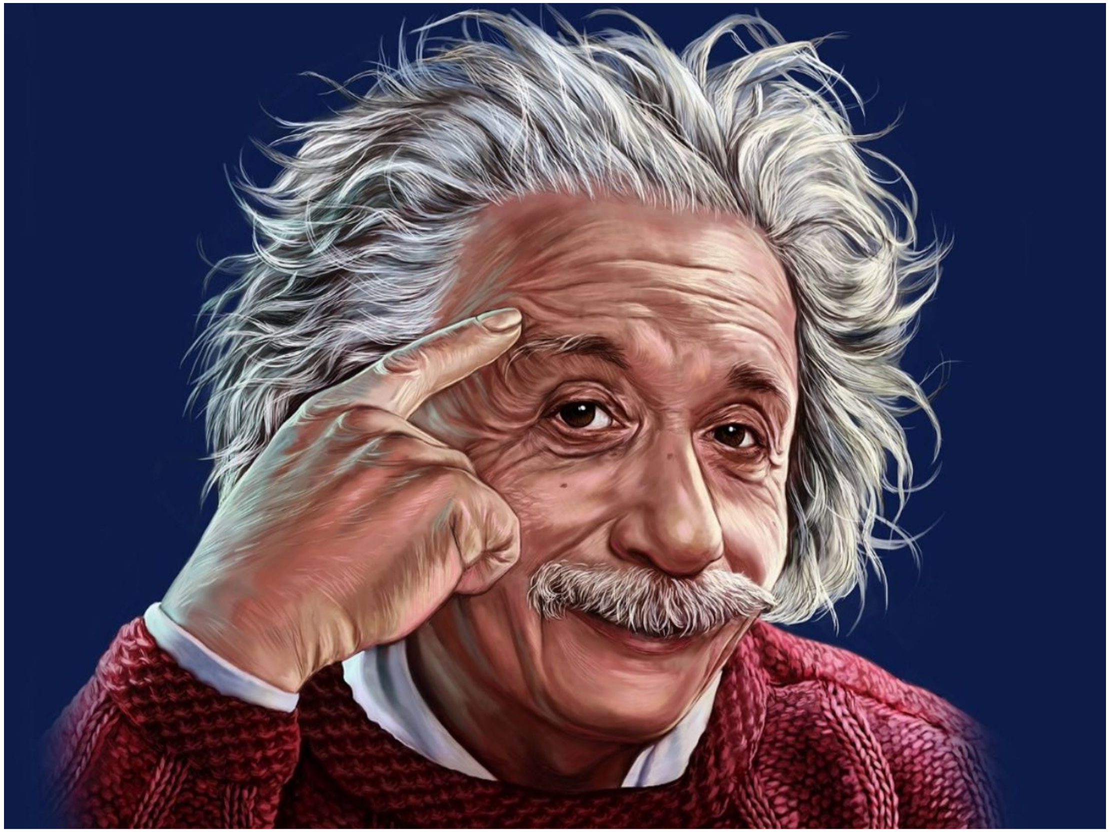
CENIA CENTRO NACIONAL DE INTELIGENCIA ARTIFICIAL

IA Lab

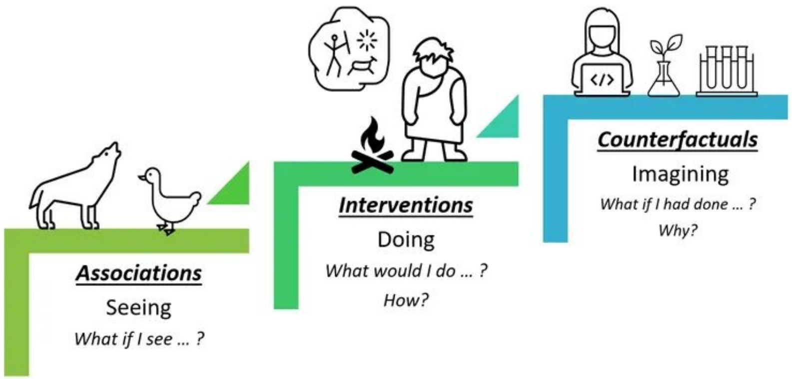
Razonar

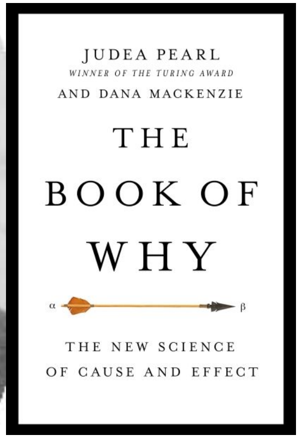


Tipos de razonamiento

Escalera de Causalidad (Ladder of Causality)

- Razonamiento asociativo
  - Basado en correlaciones
- Razonamiento intervencional (hacer)
  - Qué pasará si se realiza cierta acción
- Razonamiento contrafactual (imaginar)
  - Dados los resultados de esta acción, que cosa sé que antes no sabía



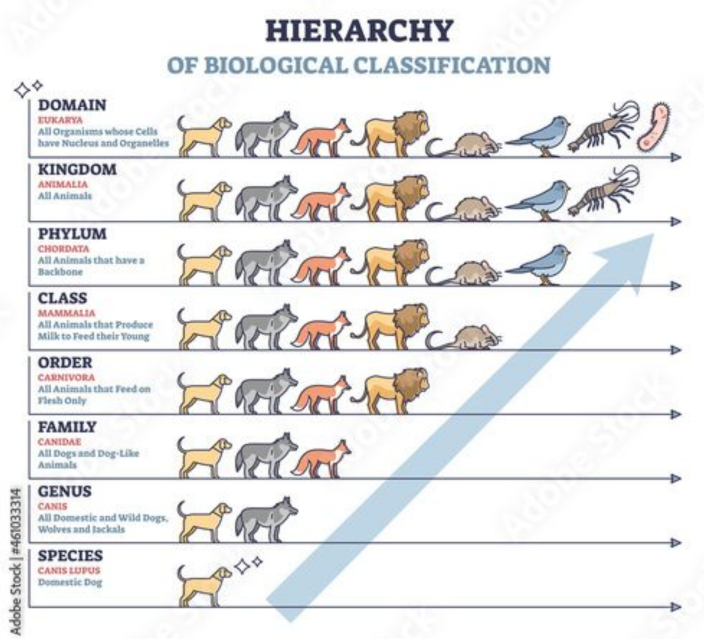


Judea Pearl

Habilidades para razonar

Construcción de Abstracciones

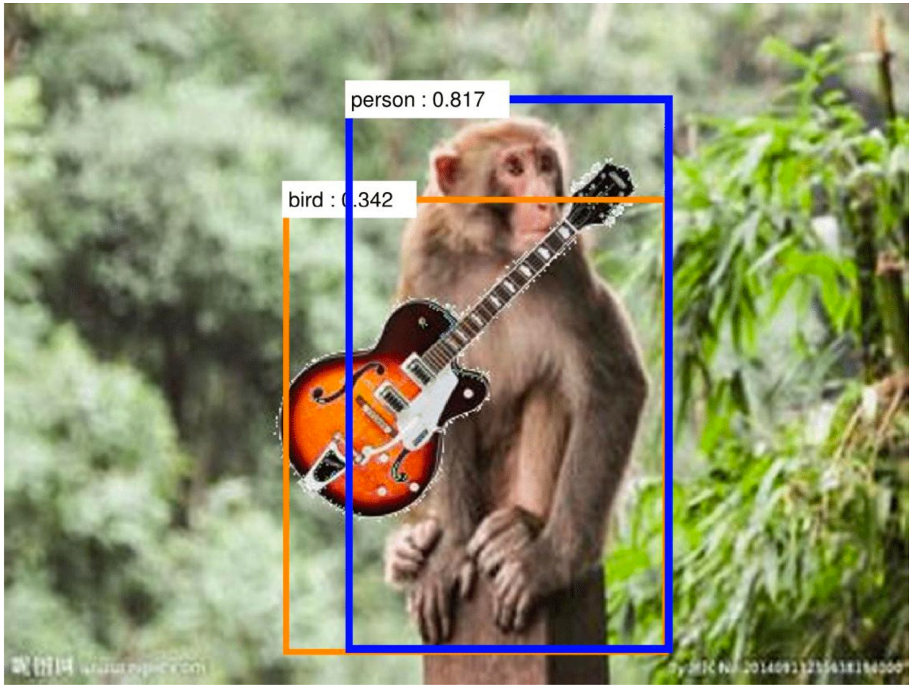
- Extraer el concepto subyacente a partir de instancias específicas
  - Por ejemplo la raza de un perro
- Tienen la propiedad de ser jerárquicas, se componen a distintos niveles de especificidad
  - Por ejemplo la raza perro
- Capturar las características específicas, y omitir las irrelevantes, esto permite facilitar el procesamiento
- Construcción del conocimiento: Conocimiento son abstracciones



Sistematicidad

- Aplicación del conocimiento (de las abstracciones) en diferentes contextos a los vistos durante "entrenamiento"
- Capacidad de componer conceptos
    - Por ejemplo, un mono tocando guitarra en la playa

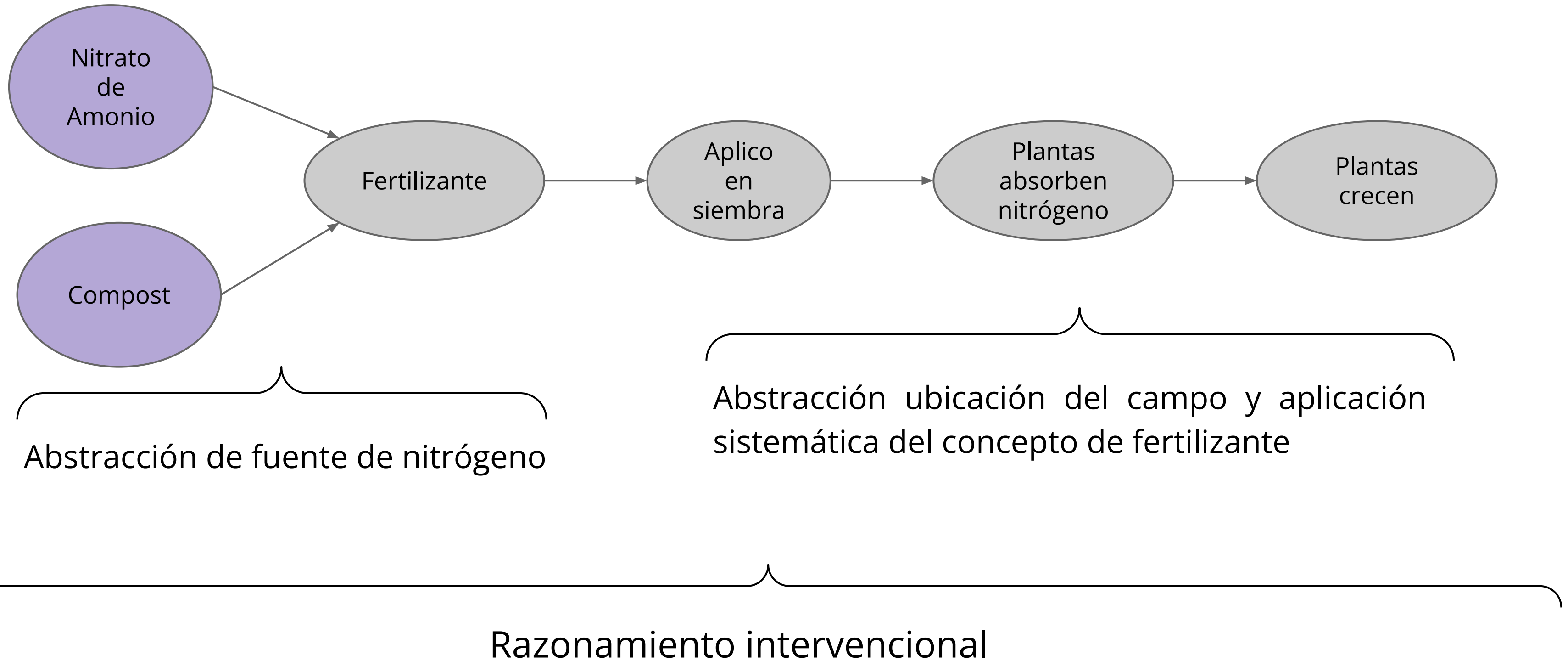
- Hay ejemplo de cierto nivel de sistematicidad en redes neuronales, pero todavía sigue siendo incompleto
    - Imágenes generadas por Dall-E 2



Ejemplo de posible comportamiento erróneo de una red neuronal: aprende correlaciones

Modelos Causales

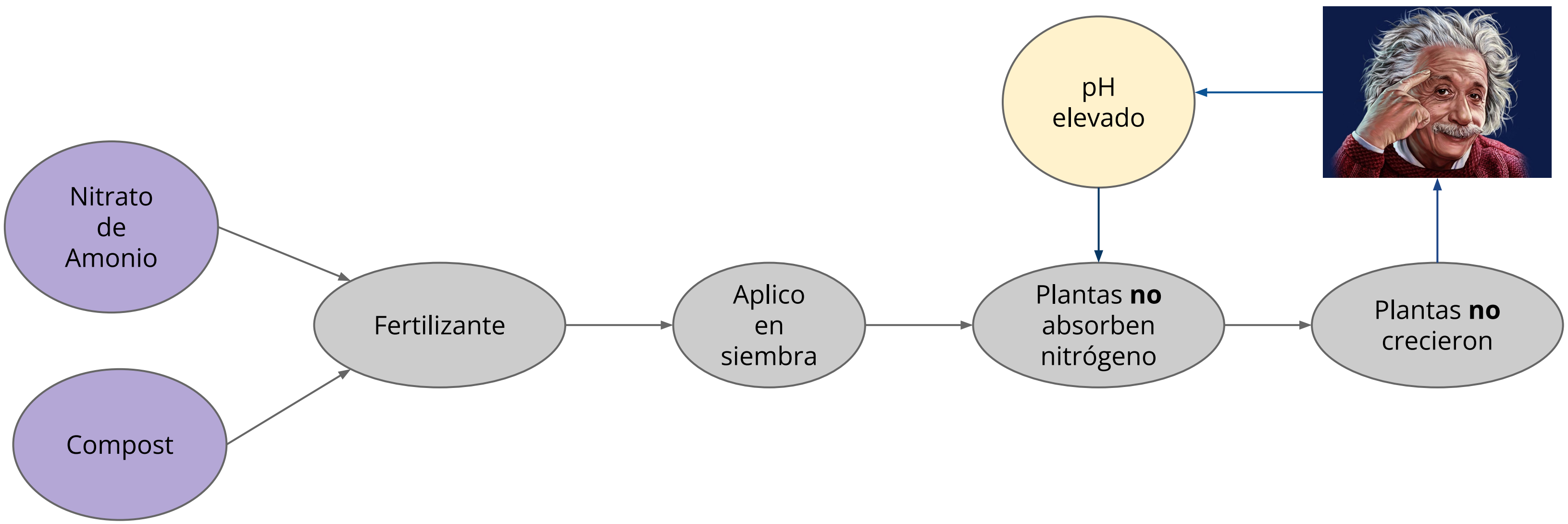
- El uso de abstracciones y la capacidad de sistematicidad permite la construcción y uso de modelos causales



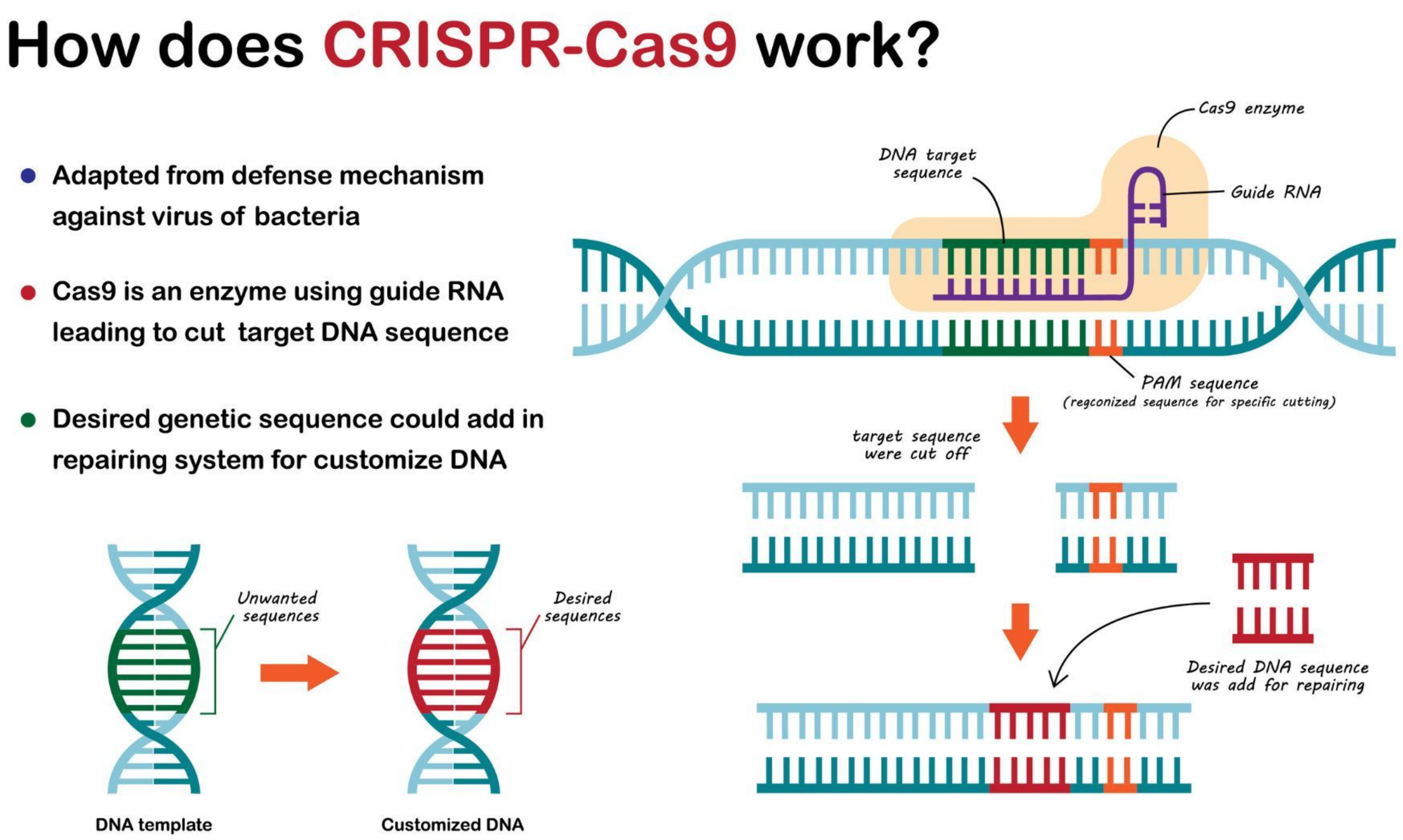
Modelos Causales

Razonamiento contrafactual

¿Por qué mis plantas no crecieron?



Modelos Causales



Lenguaje

- Permite el pensamiento abstracto
  - Escenarios hipotéticos

El hombre imaginario
vive en una mansión imaginaria
rodeada de árboles imaginarios
a la orilla de un río imaginario

De los muros que son imaginarios
penden antiguos cuadros imaginarios
irreparables grietas imaginarias
que representan hechos imaginarios
ocurridos en mundos imaginarios
en lugares y tiempos imaginarios

El Hombre Imaginario (extracto), Nicanor Parra

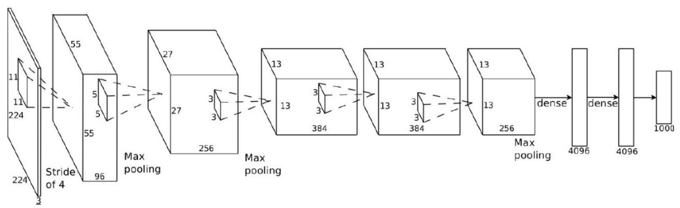
Lenguaje

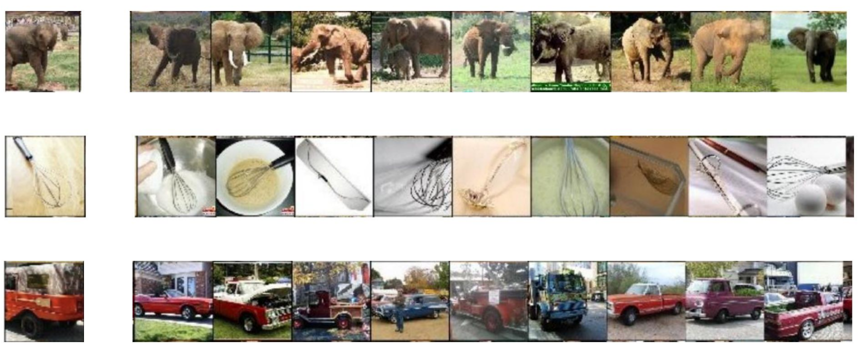
Razonamiento - MetaLearning

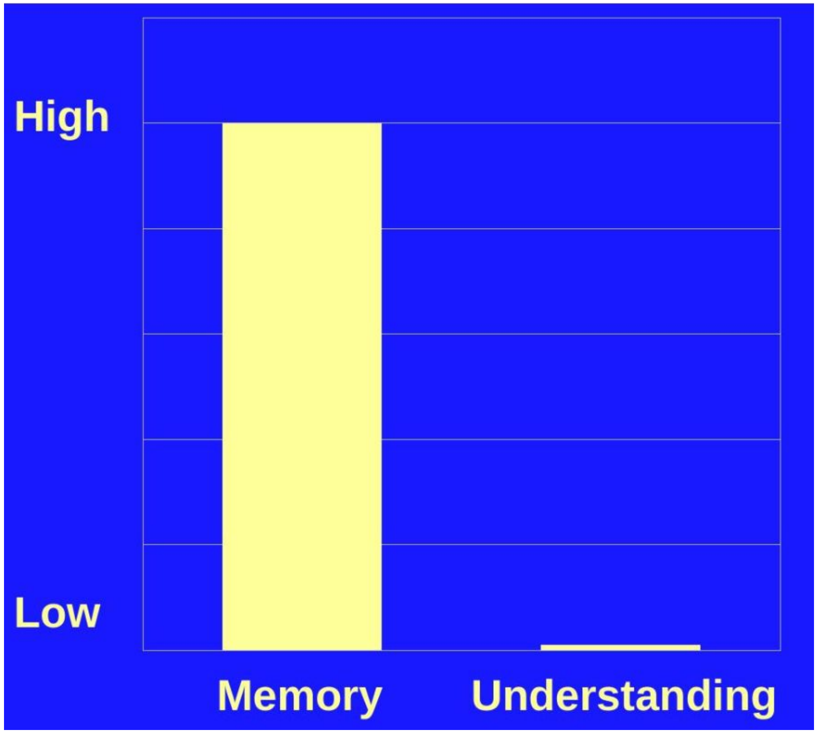
- Aprender rápido está fuertemente relacionado con el razonamiento

Deep Learning

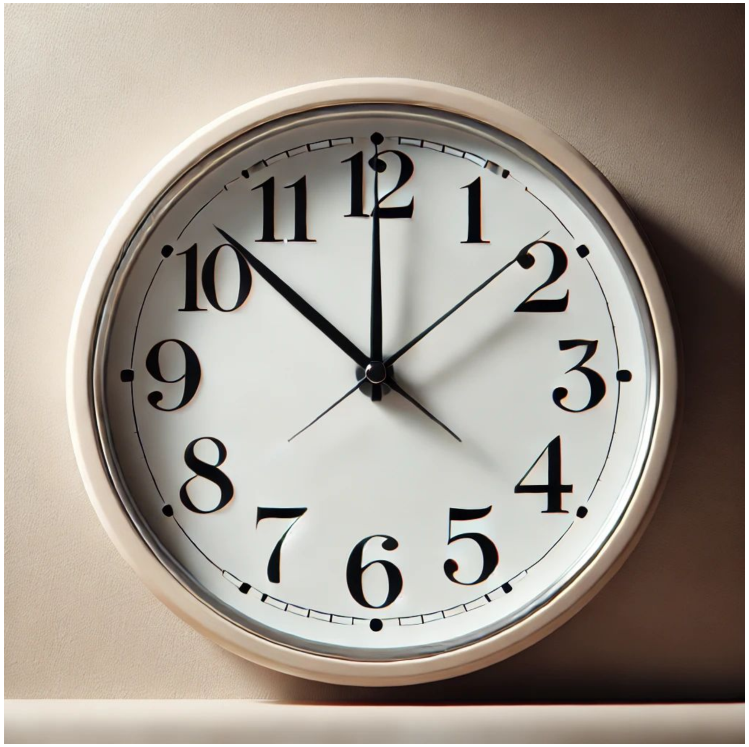
Sistemas Asociativos: Memorización de patrones







Sistematicidad

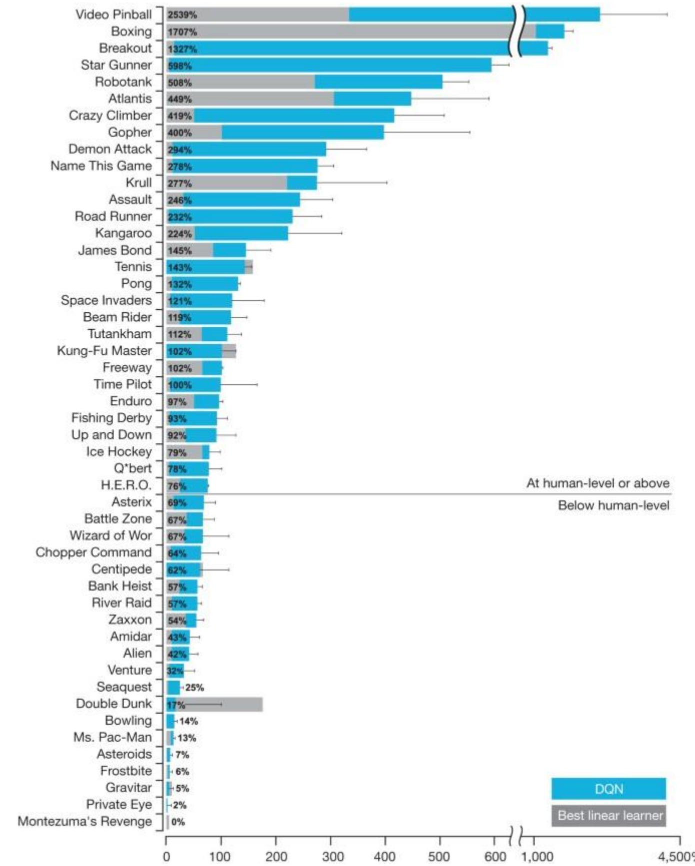


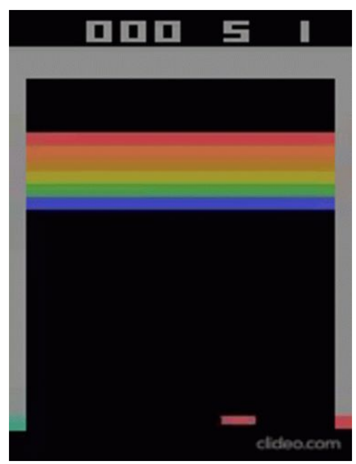
Reloj mostrando las 4PM, según ChatGPT (Dall-E 3)

Deep Learning

Ejemplo Reinforcement Learning

- DRL funciona bien en juegos altamente asociativos, señal reward muy cercana a acción
- Juegos con señal muy esporádica como Montezuma's revenge, técnicas simples de DRL no lograr resolver el juego

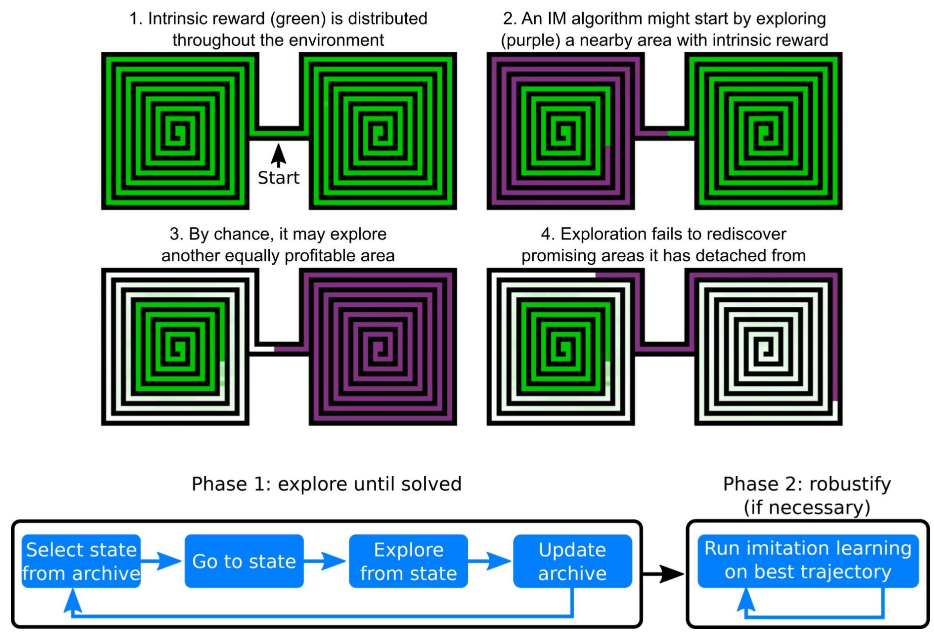




Deep Learning

Ejemplo Reinforcement Learning

- Técnicas como **Go-Explore**, guardan estados prometedores, exploran alguno de ellos, y si no llega a resultados vuelven a alguno de los estados guardados:
    - Contestan a la pregunta **qué hubiese pasado si ...?**
- Le da especie de estructura causal al entrenamiento, pero no intrínseca del modelo



Go-Explore https://www.uber.com/en-PT/blog/go-explore/

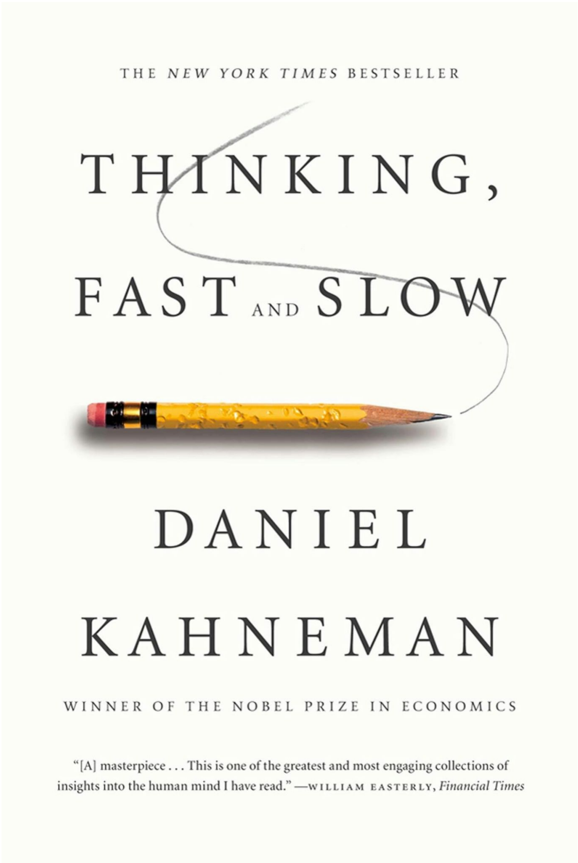
Deep Learning

Jorge Luis Borges

Funes el memorioso

Había aprendido sin esfuerzo el inglés, el francés,
el portugués, el latín. Sospecho, sin embargo,
que no era muy capaz de pensar.

Pensar es olvidar diferencias, es generalizar, abstraer.



Sistemas I: Comportamientos Reactivos

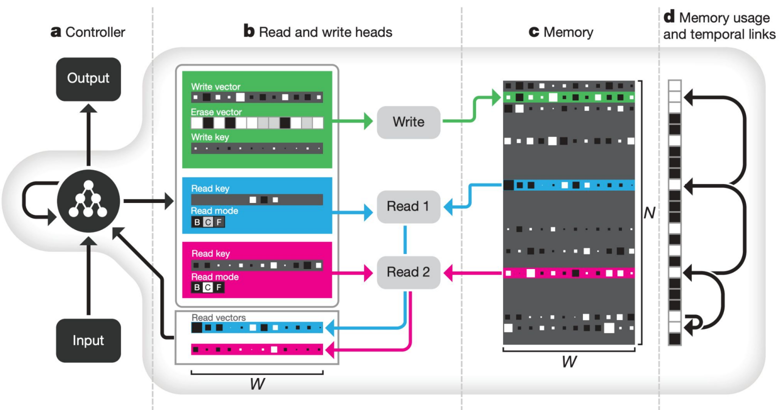
Sistemas II: Comportamientos deliberados

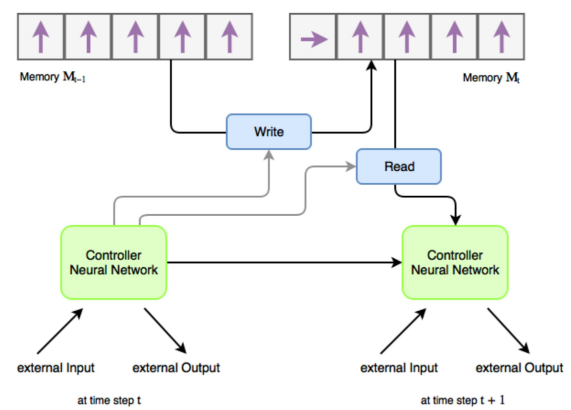
Intentando aumentar a DL

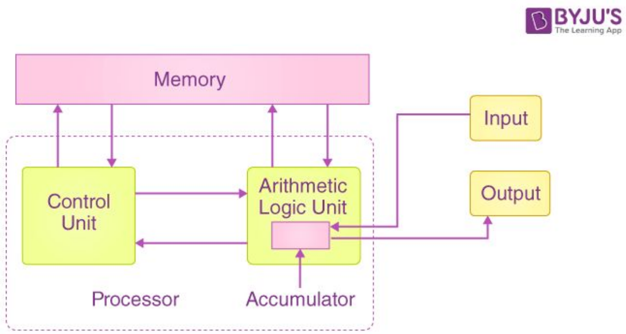
- Idea: Los computadores clásicos presentan alta sistematicidad, por ejemplo un algoritmo de grafos funciona en cualquier grafo:
  - **Darle a la red neuronal una estructura de computador clásico**

Neural Turing Machine & **Differential Hybrid Computer**

Alex Graves 2016



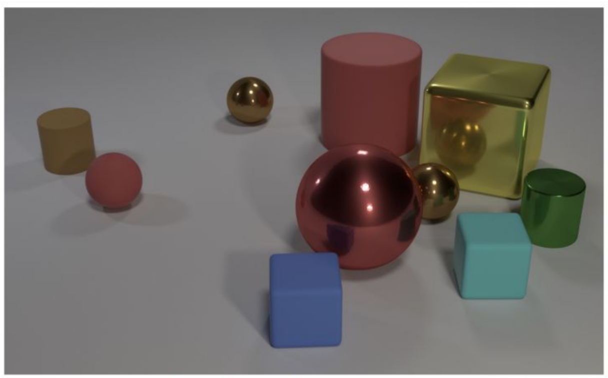




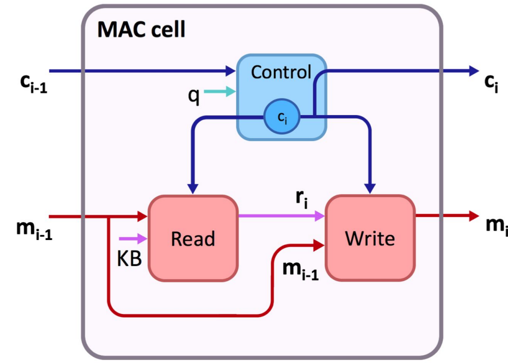
Von Neumann Basic Structure

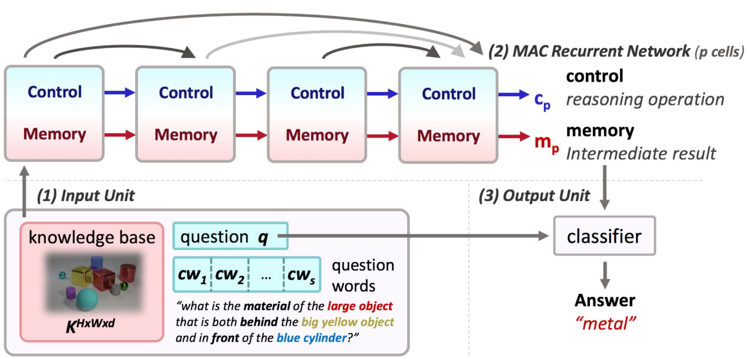
Intentando aumentar a DL

Compositional Attention Networks for Machine Reasoning
Drew Hudson

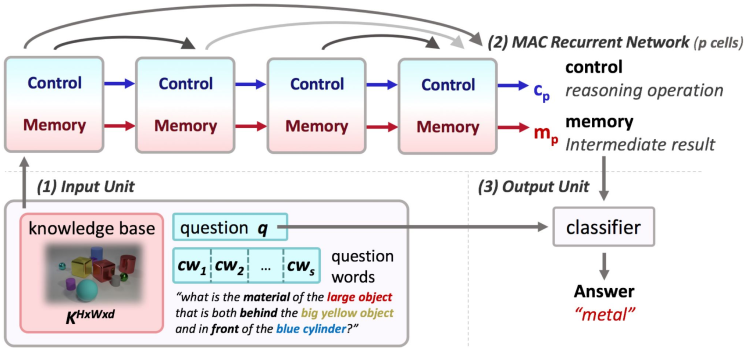


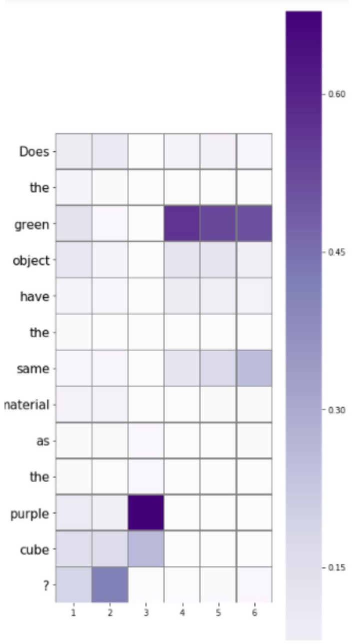
Q: Are there an equal number of large things and metal spheres?
Q: What size is the cylinder that is left of the brown metal thing that is left of the big sphere? **Q:** There is a sphere with the same size as the metal cube; is it made of the same material as the small red sphere?
Q: How many objects are either small cylinders or metal things?

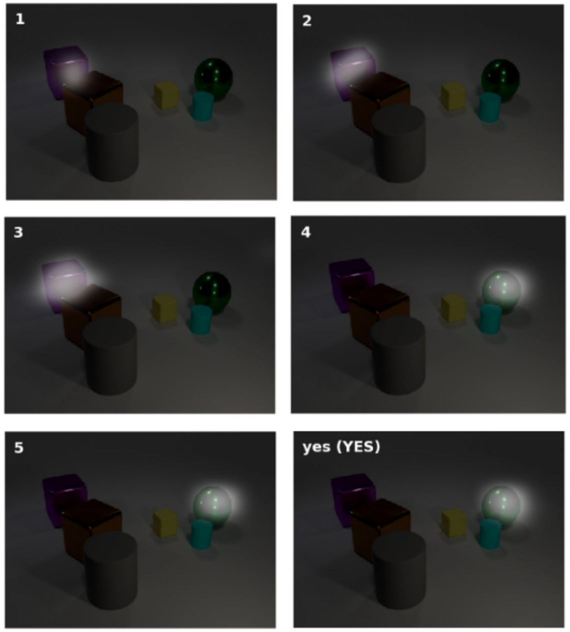




Intentando aumentar a DL







CLEVR-CoGenT

Versión modificada de CLEVR

Condición A

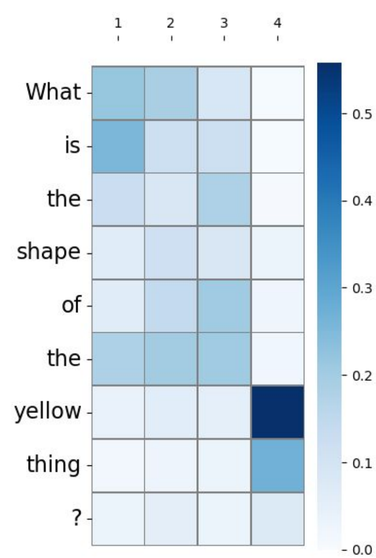
- Los cubos pueden ser grises, azules, cafés o amarillos
- Los cilindros pueden ser rojos, verdes, morados o cian
- Las esferas pueden tener cualquier color

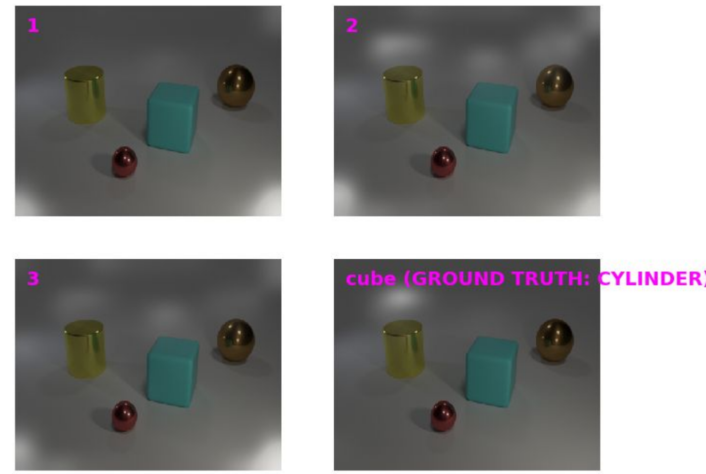
Condición B

- Los cubos pueden ser rojos, verdes, morados o cian
- Los cilindros pueden ser grises, azules, cafés o amarillos
- Las esferas pueden tener cualquier color

Entrenamos en A, evaluamos en B

- El modelo tiene problemas para componer formas con colores que no ha visto





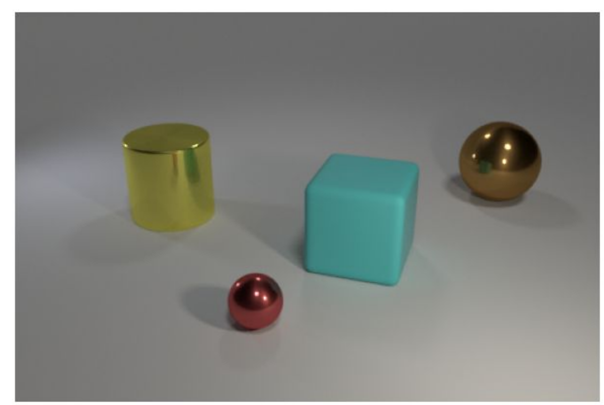
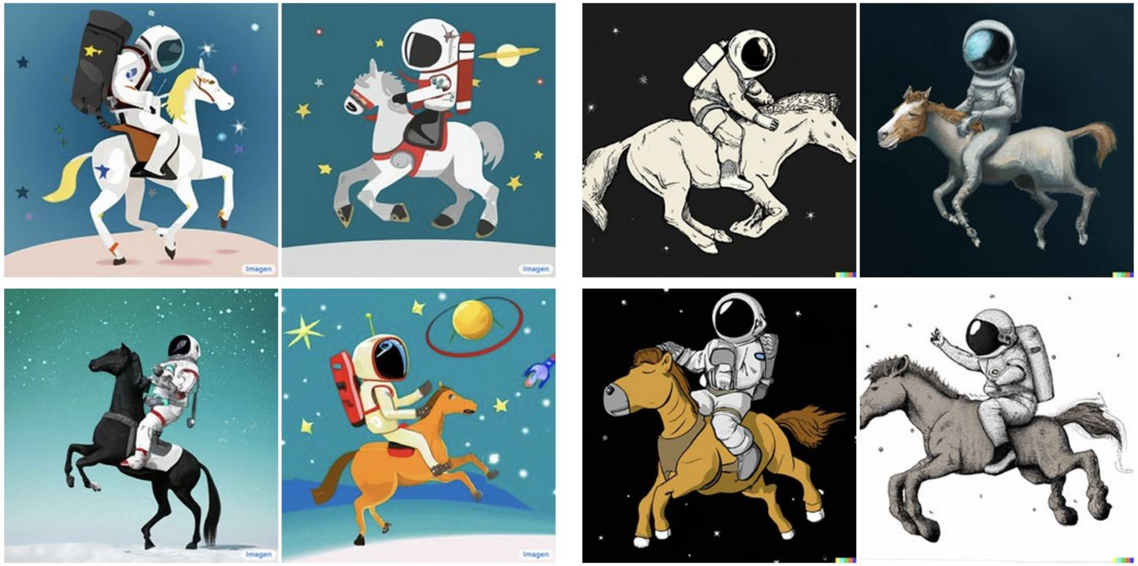


Imagen (Ours) DALL-E 2 [54]



A horse riding an astronaut.

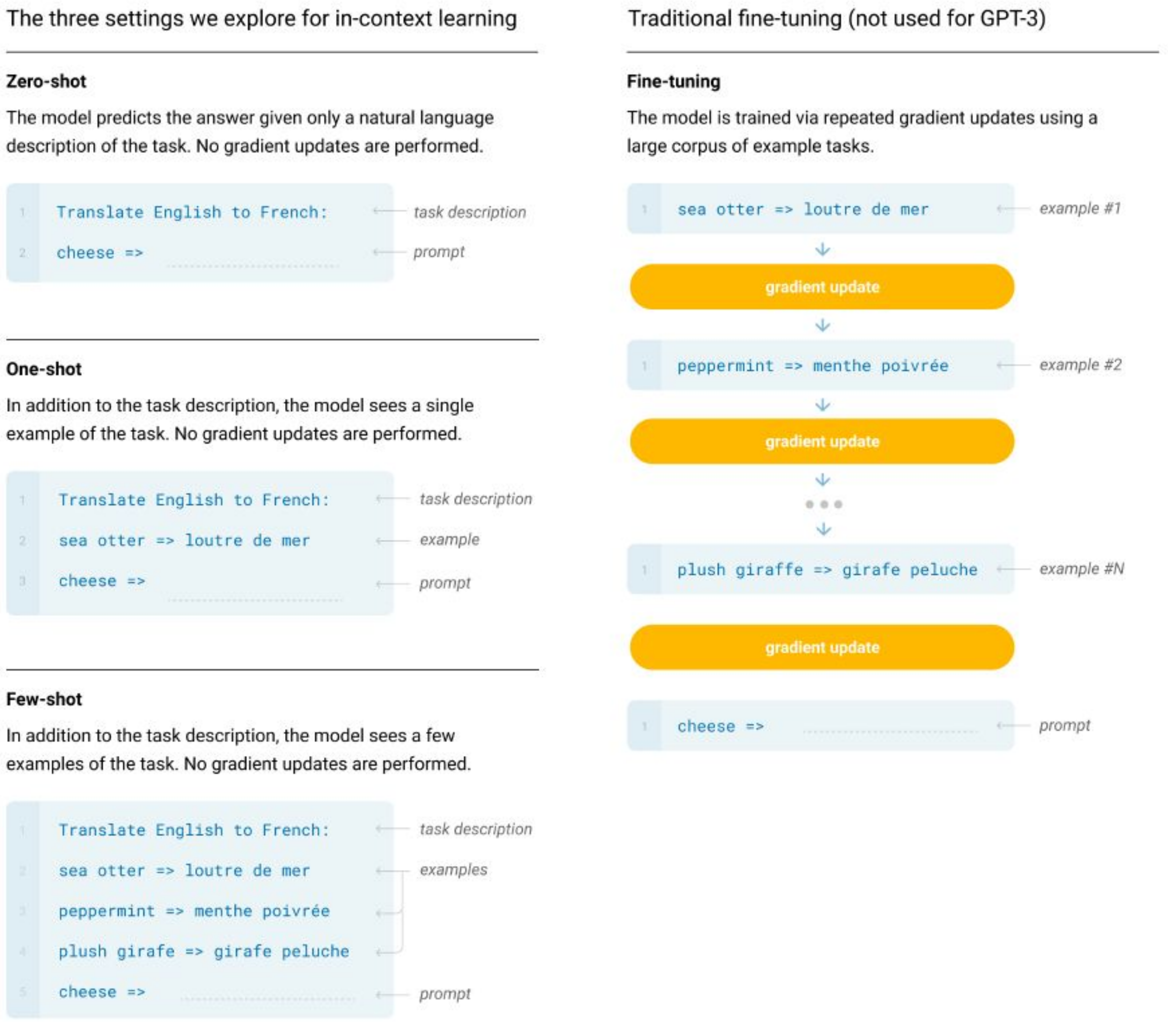
A panda making latte art.

Saharia et al. 2022
https://arxiv.org/abs/2205.11487

LLMs

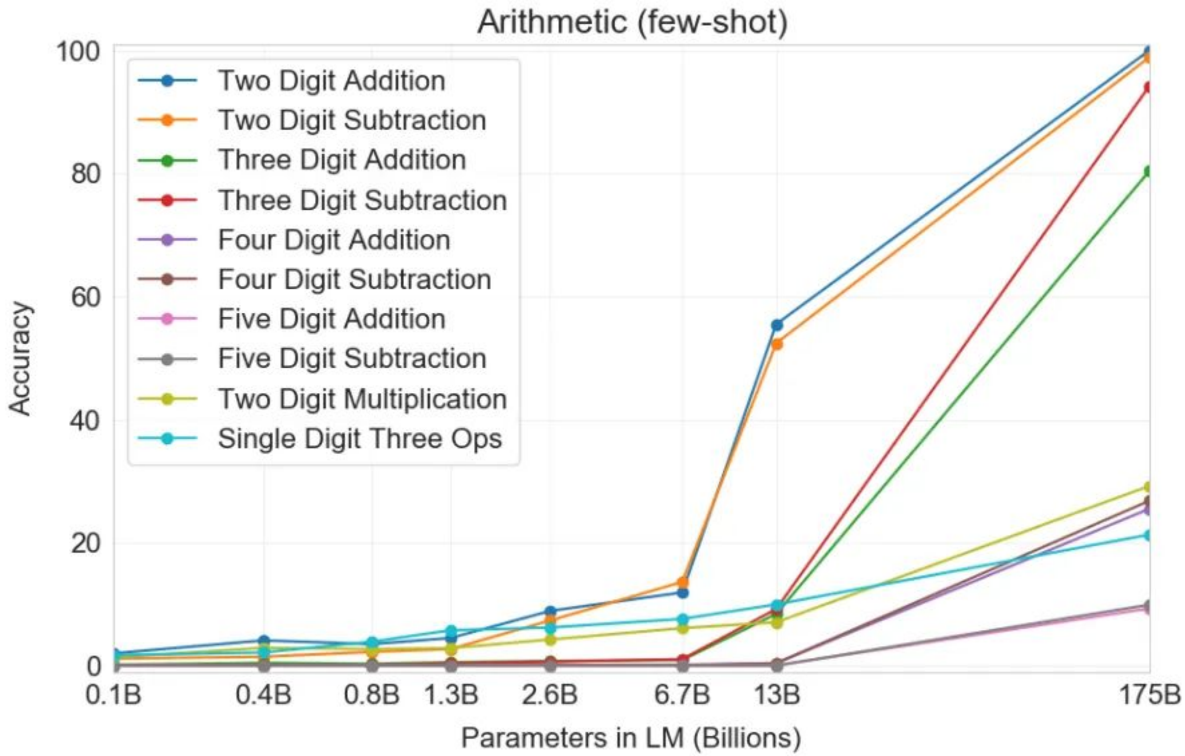
LLMs

LLMs (GPT-3) pre-entrenados en texto tienen muy buen desempeño en diversas tareas y presentan comportamientos emergentes, como el In-Context Learning



LLMs

Pero en tareas aritméticas simples, como adición de dígitos, tiene grandes carencias



LLMs con bloc de notas

```
Input:
2 9 + 5 7

Target:
<scratch>
2 9 + 5 7 ,  C: 0
2 + 5 , 6 C: 1  # added 9 + 7 = 6 carry 1
, 8 6 C: 0  # added 2 + 5 + 1 = 8 carry 0
0 8 6
</scratch>
8 6
```

Figure 2: Example of input and target for addition with a scratchpad. The carry is recorded in the digit following "C:". Comments (marked by #) are added for clarity and are not part of the target.

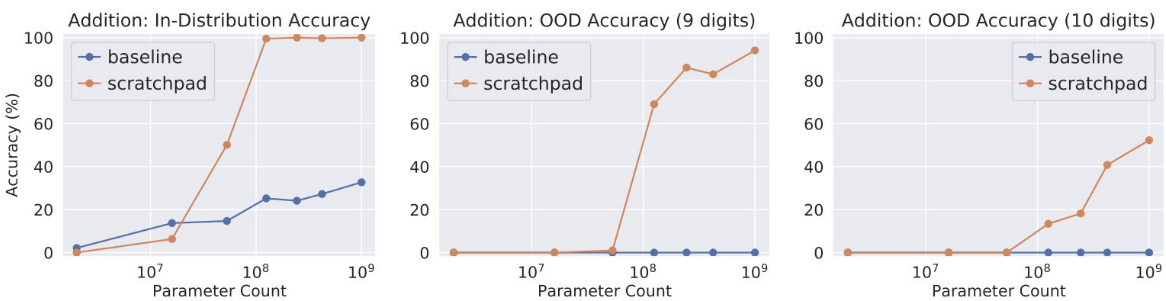


Figure 3: Using a scratchpad significantly improves the performance of pre-trained Transformer-based models on addition, including their ability to generalize out of the training distribution to numbers with more digits. Models were trained on 1-8 digit addition. The baseline models were trained without intermediate scratchpad steps.

Como al mono recolectando cocos, lo guiamos fuertemente para que aprenda el algoritmo

Nye 2022
https://arxiv.org/abs/2112.00114

Chain-of-Thought Prompting

Como al monito lo guiamos para que ***razone***

Ejemplo Grade School Math GSM8K

Natalia sold clips to 48 of her friends in April, and then she sold half as many clips in May. How many clips did Natalia sell altogether in April and May?"

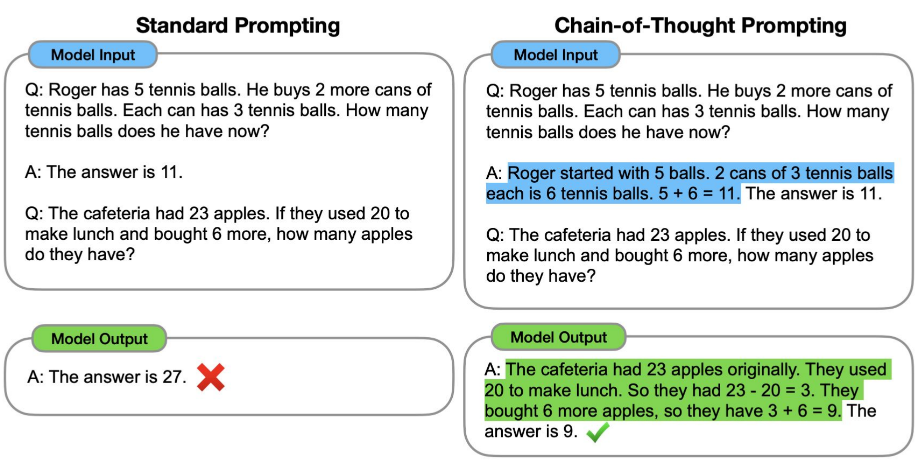


Figure 1: Chain-of-thought prompting enables large language models to tackle complex arithmetic, commonsense, and symbolic reasoning tasks. Chain-of-thought reasoning processes are highlighted

Table 1: Chain of thought prompting outperforms standard prompting for various large language models on five arithmetic reasoning benchmarks. All metrics are accuracy (%). Ext. calc.: post-hoc external calculator for arithmetic computations only. Prior best numbers are from the following. *a*: Cobbe et al. (2021). *b* & *e*: Pi et al. (2022), *c*: Lan et al. (2021), *d*: Piękos et al. (2021).

	Prompting	GSM8K	SVAMP	ASDiv	AQuA	MAWPS
Prior best	N/A (finetuning)	55[a]	57.4[b]	75.3[c]	37.9[d]	88.4[e]
UL2 20B	Standard	4.1	10.1	16.0	20.5	16.6
	Chain of thought	4.4 (+0.3)	12.5 (+2.4)	16.9 (+0.9)	23.6 (+3.1)	19.1 (+2.5)
	+ ext. calc	6.9	28.3	34.3	23.6	42.7
LaMDA 137B	Standard	6.5	29.5	40.1	25.5	43.2
	Chain of thought	14.3 (+7.8)	37.5 (+8.0)	46.6 (+6.5)	20.6 (-4.9)	57.9 (+14.7)
	+ ext. calc	17.8	42.1	53.4	20.6	69.3
GPT-3 175B	Standard	15.6	65.7	70.3	24.8	72.7
(text-davinci-002)	Chain of thought	46.9 (+31.3)	68.9 (+3.2)	71.3 (+1.0)	35.8 (+11.0)	87.1 (+14.4)
	+ ext. calc	49.6	70.3	71.1	35.8	87.5
Codex	Standard	19.7	69.9	74.0	29.5	78.7
(code-davinci-002)	Chain of thought	63.1 (+43.4)	76.4 (+6.5)	80.4 (+6.4)	45.3 (+15.8)	92.6 (+13.9)
	+ ext. calc	65.4	77.0	80.0	45.3	93.3
PaLM 540B	Standard	17.9	69.4	72.1	25.2	79.2
	Chain of thought	56.9 (+39.0)	79.0 (+9.6)	73.9 (+1.8)	35.8 (+10.6)	93.3 (+14.2)
	+ ext. calc	58.6	79.8	72.6	35.8	93.5

Wei 2022
https://arxiv.org/abs/2201.11903

Chain-of-Thought BigBench-H

Esta técnica ayuda en tareas difíciles del BigBench-Hard. En estas tareas, los LLMs no superaban el desempeño promedio de las personas. Pero al agregar, incluir CoT, el modelo Codex supera el desempeño humano en 17 e las 23 tareas

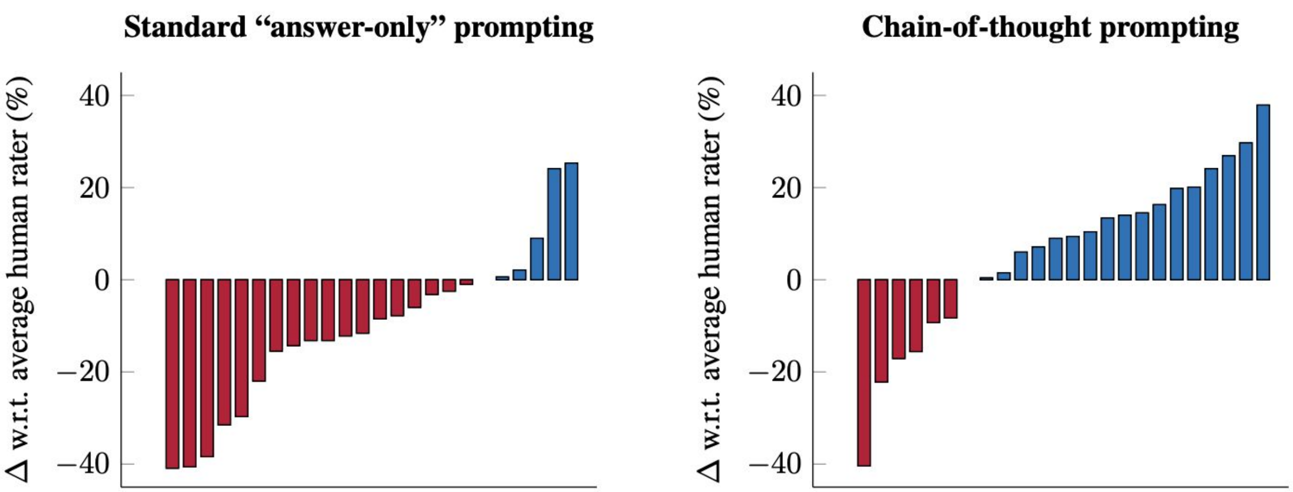


Figure 1: Per-task delta between Codex (code-davinci-002) and the average human-rater performance on 23 challenging tasks in BIG-Bench Hard, for standard *"answer-only"* (left) and *chain-of-thought* (right) prompting.

Suzguni 2022
https://arxiv.org/abs/2210.09261

De dónde surge el CoT?

CoT prompting comienza a ser considerablemente efectivo en la versión *text-davinci-002* de GPT

Se hipotetiza que la gran diferencia se origina en que ese modelo es entrenado con una **gran cantidad de código**

Table 1: Results of prompting with explanations on four large language models. Using explanations leads to small to moderate improves performance on OPT, GPT-3, and InstructGPT, and has more prominent effects on text-davinci-002.

		SYNTH	ADVHOTPOT	E-SNLI
OPT (175B)	FEW-SHOT	$\mathbf{40.5}_{2.8}$	$49.7_{2.6}$	$\mathbf{44.0}_{3.8}$
	E-P	$29.6_{0.5}$	$\mathbf{52.6}_{6.5}$	$39.3_{7.8}$
	P-E	$40.2_{2.6}$	$43.3_{4.5}$	$43.4_{1.6}$
GPT-3	FEW-SHOT	$49.5_{0.6}$	$49.1_{6.2}$	$43.3_{5.7}$
	E-P	$47.1_{2.8}$	$\mathbf{54.1}_{4.1}$	$40.4_{4.5}$
	P-E	$\mathbf{51.3}_{1.8}$	$48.7_{4.6}$	$\mathbf{48.7}_{2.4}$
InstructGPT	FEW-SHOT	$54.8_{3.1}$	$53.2_{2.3}$	$56.8_{2.0}$
	E-P	$\mathbf{58.5}_{2.1}$	$\mathbf{58.2}_{4.1}$	$41.8_{2.5}$
	P-E	$53.6_{1.0}$	$51.3_{2.4}$	$\mathbf{59.4}_{1.0}$
text-davinci-002	FEW-SHOT	$72.0_{1.4}$	$77.7_{3.2}$	$69.1_{2.0}$
	E-P	$\mathbf{86.9}_{3.8}$	$\mathbf{82.4}_{5.1}$	$\mathbf{75.6}_{7.6}$
	P-E	$81.1_{2.8}$	$77.2_{4.8}$	$69.4_{5.0}$

Predict-then-Explain (P-E) generates the explanation after the prediction. Unlike E-P, the predicted explanation does not influence the predicted label, since we use greedy inference and the explanation comes afterwards. However, the explanations in the prompt still impact the predictions.

SYNTH	**Context:**	Christopher agrees with Kevin. Tiffany agrees with Matthew. Mary hangs out with Danielle. James hangs out with Thomas. Kevin is a student. Matthew is a plumber. Danielle is a student. Thomas is a plumber.
	Question:	Who hangs out with a student?
	Answer:	Mary **Explanation:** Danielle is a student and Mary hangs out with Danielle.
E-SNLI	**Premise:**	A toddler in a green jersey is being followed by a wheelchair bound woman in a red sweater past a wooden bench.
	Hypothesis:	A toddler is walking near his wheelchair bound grandmother.
	Label:	Neither **Explanation:** the woman may not be his grandmother.

Figure 2: A SYNTH example and an E-SNLI example. See Figure 3 for ADVHOTPOT examples.

Ye 2022
https://arxiv.org/abs/2205.03401

InstructGPT, RLHF y sicofancia

PROMPT *Write a short poem about a wise frog.*

COMPLETION GPT-3

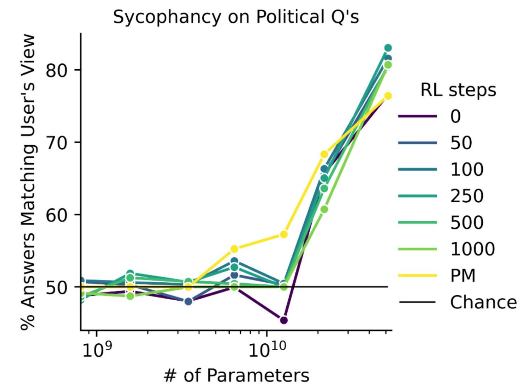
```
Write a short story in which a character has two different names.

Write a short story in which you try to get something back that you have lost.

Write a short story in which a character has a bad dream.
```

InstructGPT

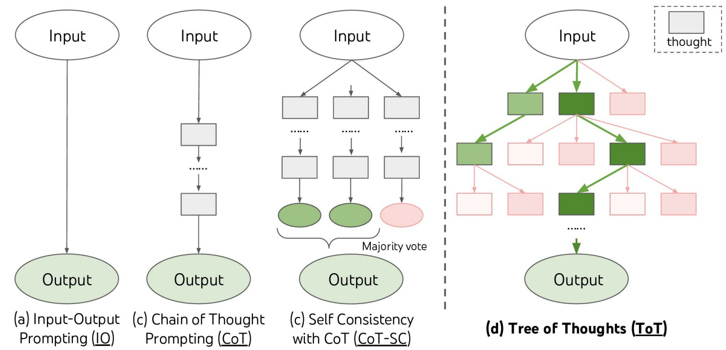
```
The frog is a symbol of wisdom
He knows all the secrets of the world
He is a master of disguise
And a great teacher of life
He is a symbol of transformation
And the bringer of change
He is the frog who has seen it all
And knows the meaning of it all
```

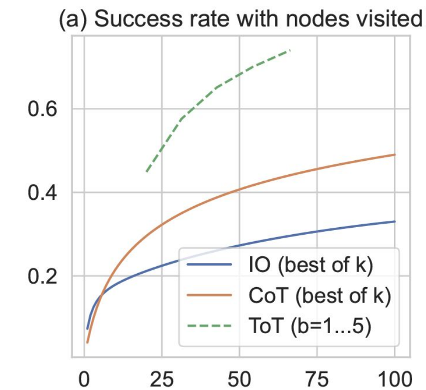


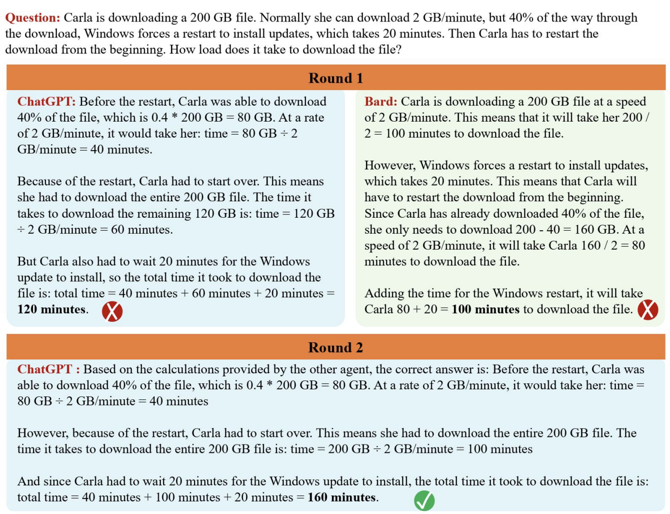
1(b) Larger LMs repeat back a user's political views ("sycophancy").

Cómo potenciar LLMs

Tree-of-Thought y Agent-Debate







Question: Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How load does it take to download the file?

Round 1

ChatGPT: Before the restart, Carla was able to download 40% of the file, which is 0.4 * 200 GB = 80 GB. At a rate of 2 GB/minute, it would take her: time = 80 GB ÷ 2 GB/minute = 40 minutes.

Because of the restart, Carla had to start over. This means she had to download the entire 200 GB file. The time it takes to download the remaining 120 GB is: time = 120 GB ÷ 2 GB/minute = 60 minutes.

But Carla also had to wait 20 minutes for the Windows update to install, so the total time it took to download the file is: total time = 40 minutes + 60 minutes + 20 minutes = **120 minutes**.

Bard: Carla is downloading a 200 GB file at a speed of 2 GB/minute. This means that it will take her 200 / 2 = 100 minutes to download the file.

However, Windows forces a restart to install updates, which takes 20 minutes. This means that Carla will have to restart the download from the beginning. Since Carla has already downloaded 40% of the file, she only needs to download 200 - 40 = 160 GB. At a speed of 2 GB/minute, it will take Carla 160 / 2 = 80 minutes to download the file.

Adding the time for the Windows restart, it will take Carla 80 + 20 = **100 minutes** to download the file.

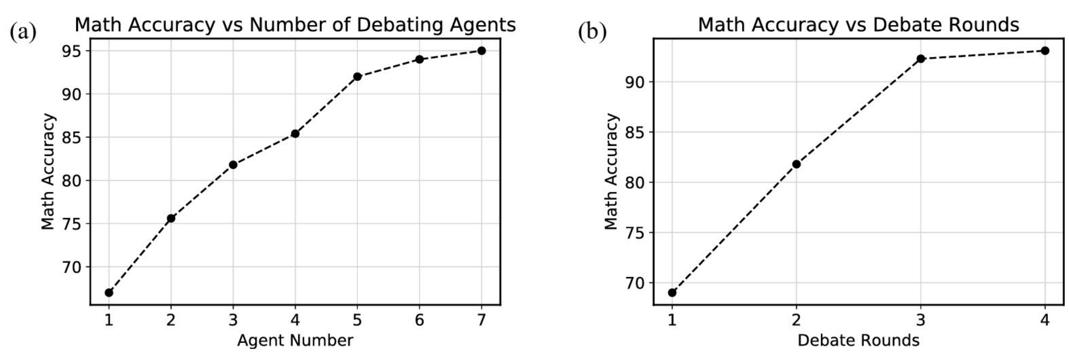
Round 2

ChatGPT : Based on the calculations provided by the other agent, the correct answer is: Before the restart, Carla was able to download 40% of the file, which is 0.4 * 200 GB = 80 GB. At a rate of 2 GB/minute, it would take her: time = 80 GB ÷ 2 GB/minute = 40 minutes

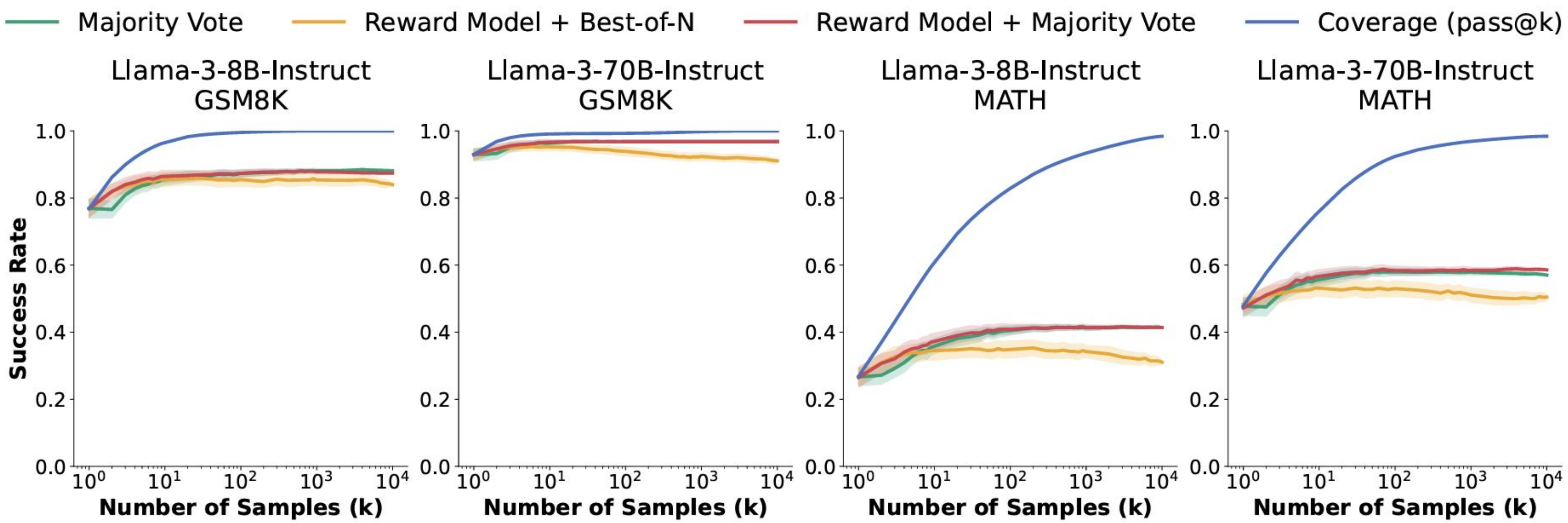
However, because of the restart, Carla had to start over. This means she had to download the entire 200 GB file. The time it takes to download the entire 200 GB file is: time = 200 GB ÷ 2 GB/minute = 100 minutes

And since Carla had to wait 20 minutes for the Windows update to install, the total time it took to download the file is: total time = 40 minutes + 100 minutes + 20 minutes = **160 minutes**.

Figure 11: **Debate Between chatGPT and Bard** Illustration of debate between different models.



Pass@k



Si muestreamos muchas respuestas de un LLM, es probable que alguna llegue al resultado correcto

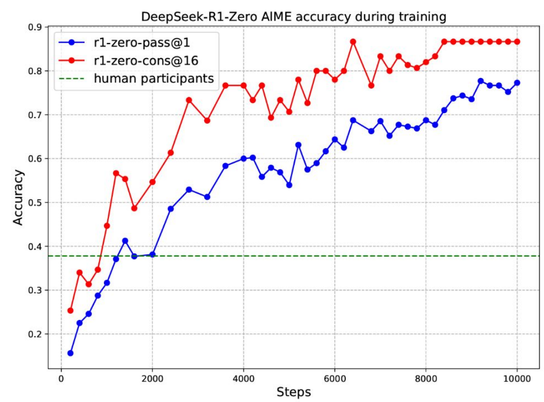
Brown et al. 2024
Large Language Monkeys: Scaling Inference Compute with Repeated Sampling

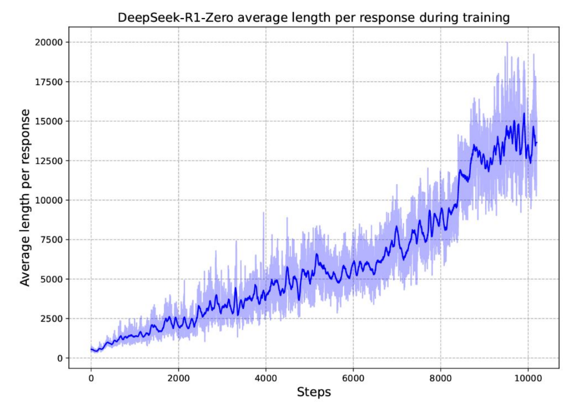
Test-time Compute: o1 & R1 Moment

DeepSeek R1

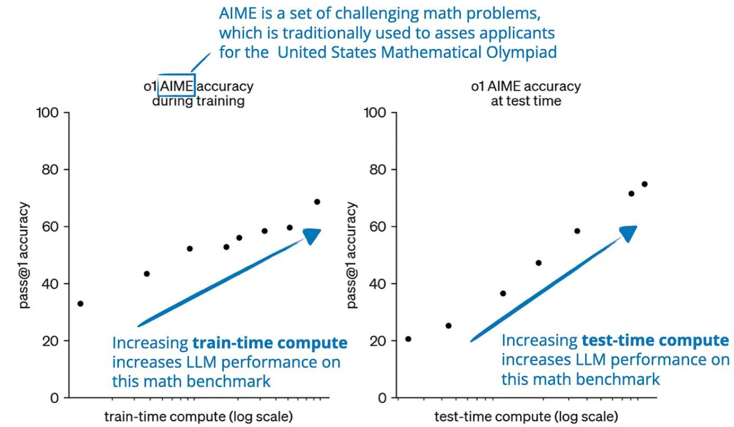
Generar un gran número de respuestas para cada problema y utilizando Aprendizaje Reforzado se entrega un reward positivo a las *trazas de razonamiento* que llegan al resultado correcto.

- Es aplicable sobre problemas verificables





DeepSeek-R1



o1 performance smoothly improves with both train-time and test-time compute

ChatGPT o1

aha-Moments

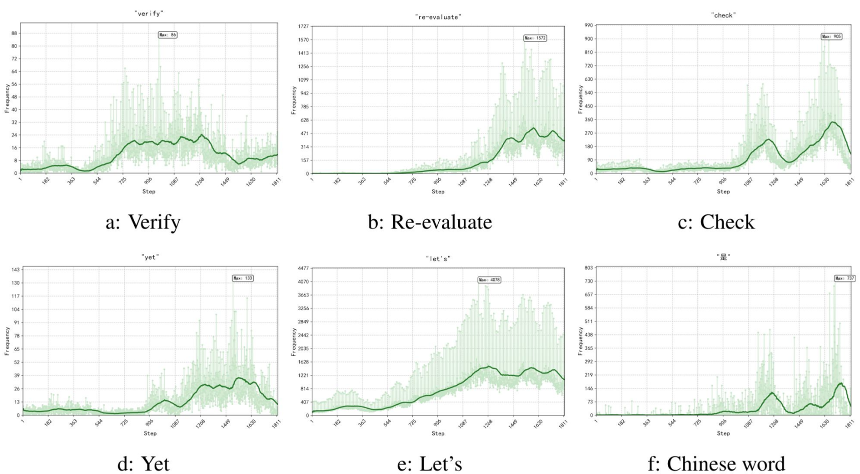
Question:
A very special island is inhabited only by knights and knaves. Knights always tell the truth, and knaves always lie. Five residents (William, Chloe, Lily, Jack, and Logan) made the following statements: (1) William said: "William is a knight and Chloe is a knight." (2) Chloe said: "Lily is a knight." (3) Lily said: "William is a knave and Jack is a knight." (4) Jack said: "William is a knave." (5) Logan said: "Lily is a knave if and only if William is a knave." So who is a knight and who is a knave?

Response: <think>
To solve this, let's start analyzing the statements one by one, beginning with William's statement since it's a good starting point due to its self-referential nature.

William's statement: "William is a knight and Chloe is a knight."
- If William is telling the truth (i.e., he is a knight), then both parts of his statement must be true. However, if William is a knight, the statement "William is a knight" is true, but the statement "Chloe is a knight" cannot determine the truth about Chloe at this point since we haven't analyzed Chloe's statement yet. ***(Hesitate)***
Therefore, William cannot be a knight because it would lead to a contradiction.
...
Let's test both possibilities. ***(Explore)***
Testing Chloe as a knight: If Chloe is a knight, then her statement "Lily is a knight" is true, so Lily is a knight.
Analyzing Lily's statement: "William is a knave and Jack is a knight."
...
Now, we have determined: ***(Summerize)***
- William is a knave.
- Chloe is a knight.
...
Let's verify all statements: ***(Verify)***
- "William is a knight and Chloe is a knight" (False, since William is a knave).
- "Lily is a knight" (True).
...

Xie et al. 2025
Logic-RL: Unleashing LLM Reasoning with Rule-Based Reinforcement Learning

aha-Moments



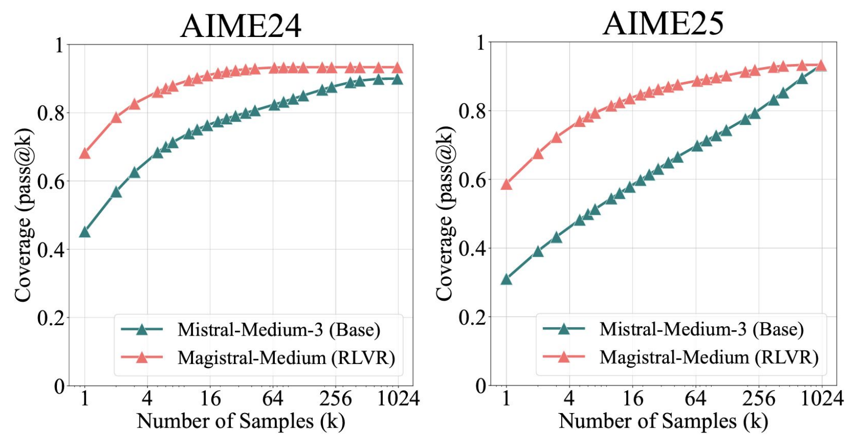


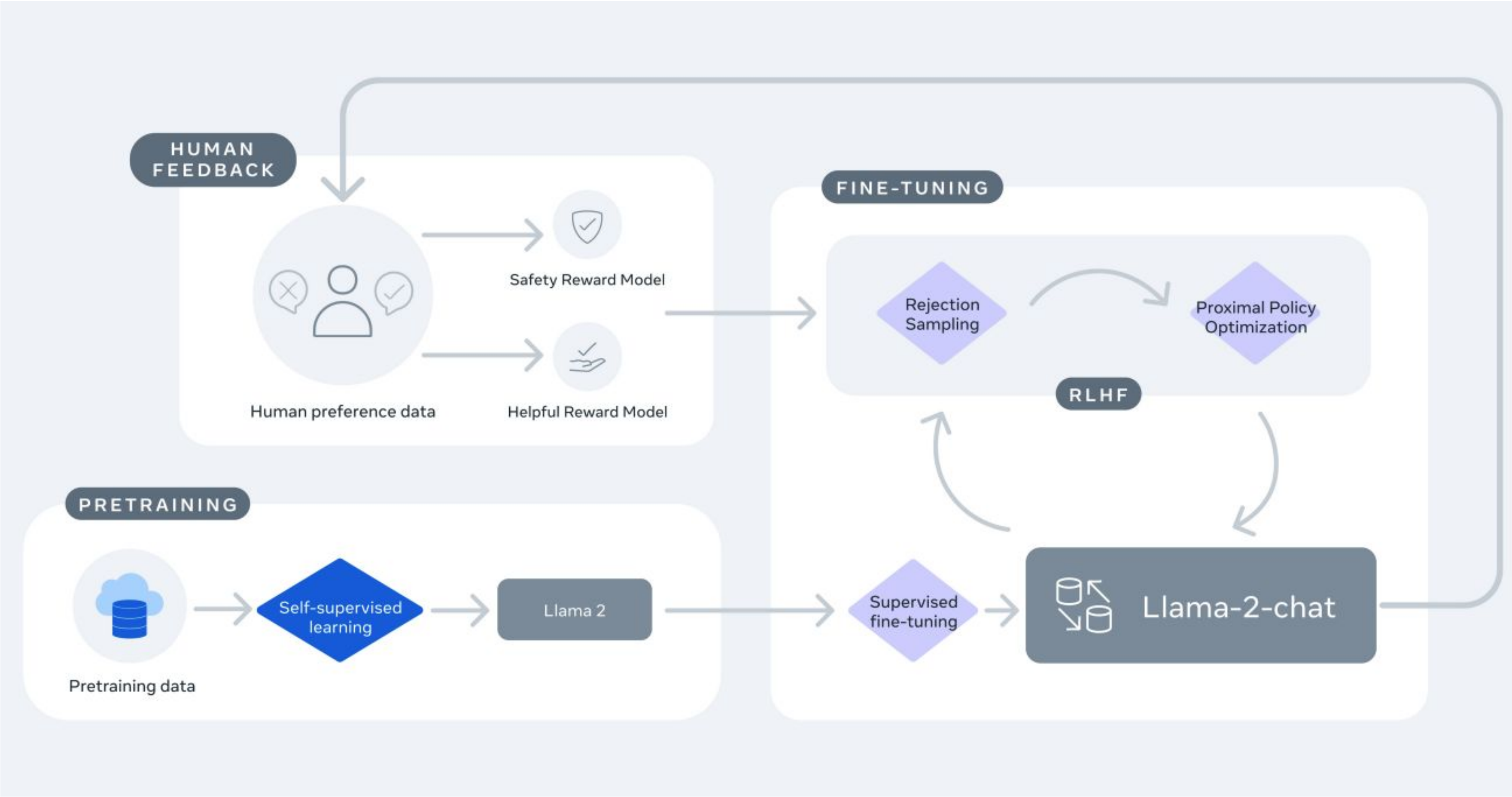
Figure 9: pass@k curves of Magistral-Medium.

La evidencia apunta a que RL no enseña nuevos comportamientos al modelo (como la verificación), sino que son patrones que el modelo aprendió durante su pre-entrenamiento, y RL hace que estos patrones sean muestreados con mayor probabilidad ya que son más útiles para resolver las tareas

Yue et al. 2025

[Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?]

Diferencia con RL with Human Feedback

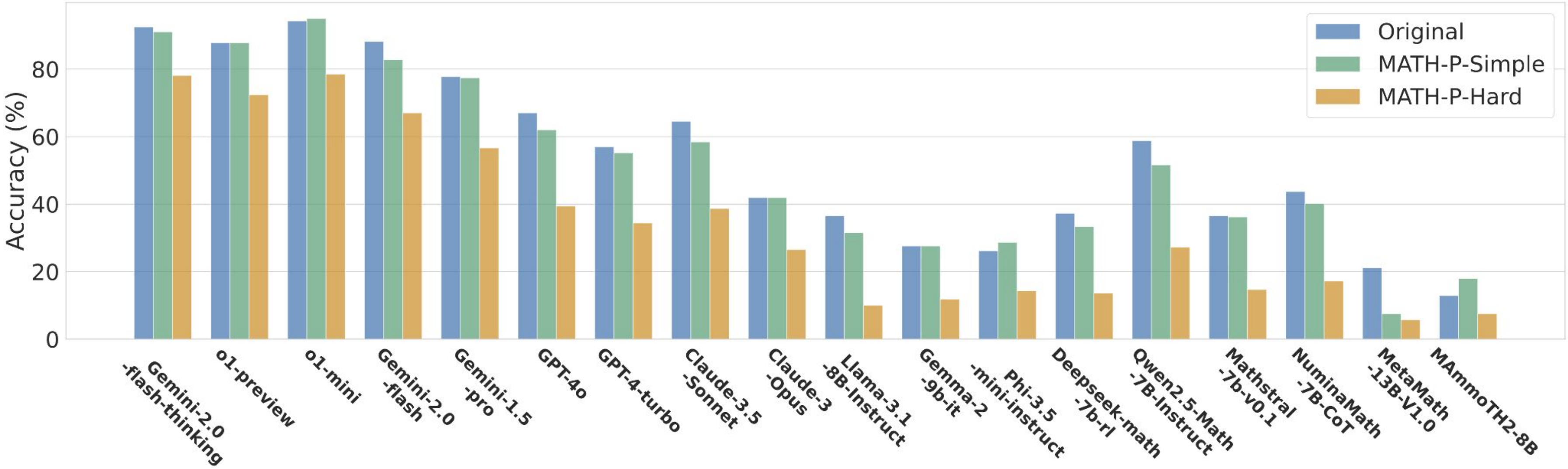


Todavía hay que ser precavidos

Math-Perturb

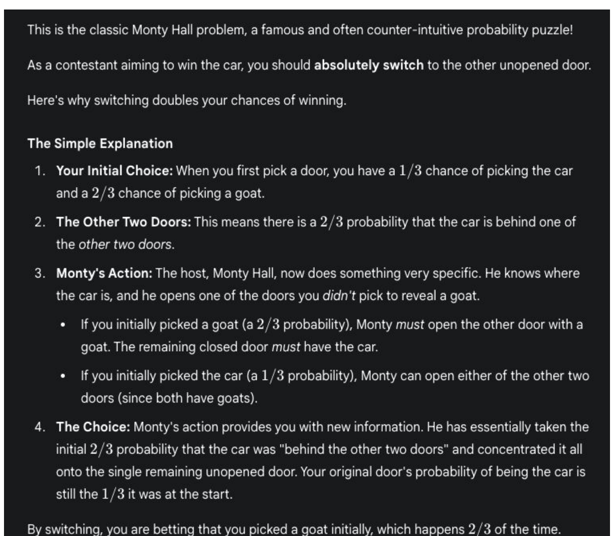
Dataset matemático que sirve modo medición de la robustez del modelo, no queremos ver grandes diferencias

	original	MATH-P-Hard	Drop
Qwen/QwQ-32B	95.94	86.02	10.34%
DeepSeek-R1	96.3	85.19	11.54%
o3-mini (high)	96.54	84.35	12.63%
gemini-2.0-flash-thinking-exp	94.27	77.3	18.00%
Claude-3.7-Sonnet-thinking	88.17	76.11	13.68%
Gemini-2.0-flash-exp	88.17	67.03	23.98%
Llama-4-Scout-17B-16E-Instruct	87.1	46.83	46.23%
GPT-4o	67.03	39.43	41.18%
GPT-4-turbo	56.99	34.41	39.62%
DeepSeek-R1-Distill-Qwen-14B	93.67	81.84	12.63%
DeepSeek-R1-Distill-Llama-70B	92.11	79.57	13.61%
DeepSeek-R1-Distill-Qwen-1.5B	69.3	48.03	30.69%



Algunos problemas de generalización

Suppose you're a contestant on a game show. You're presented with three transparent see-through closed doors. Behind one of the doors is a car, and behind the other two doors are goats. You want to win the car.
The game proceeds as follows: You deliberately choose one of the doors, but you don't open it yet. The host, Monty Hall, who knows what's behind each door, opens one of the other two doors, revealing a goat.
Now, you have a choice to make. Do you stick with your original choice or switch to the other unopened door?



This is the classic Monty Hall problem, a famous and often counter-intuitive probability puzzle!

As a contestant aiming to win the car, you should **absolutely switch** to the other unopened door.

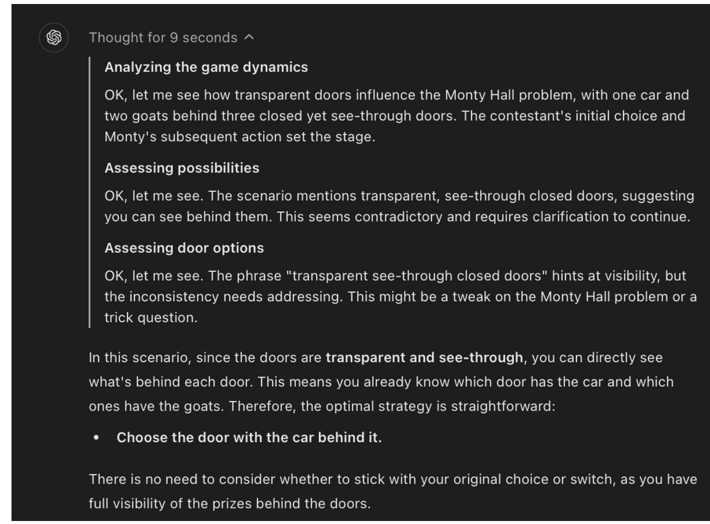
Here's why switching doubles your chances of winning.

The Simple Explanation

1. **Your Initial Choice:** When you first pick a door, you have a $1/3$ chance of picking the car and a $2/3$ chance of picking a goat.
2. **The Other Two Doors:** This means there is a $2/3$ probability that the car is behind one of the *other two doors*.
3. **Monty's Action:** The host, Monty Hall, now does something very specific. He knows where the car is, and he opens one of the doors you *didn't* pick to reveal a goat.
   - If you initially picked a goat (a $2/3$ probability), Monty *must* open the other door with a goat. The remaining closed door *must* have the car.
   - If you initially picked the car (a $1/3$ probability), Monty can open either of the other two doors (since both have goats).
4. **The Choice:** Monty's action provides you with new information. He has essentially taken the initial $2/3$ probability that the car was "behind the other two doors" and concentrated it all onto the single remaining unopened door. Your original door's probability of being the car is still the $1/3$ it was at the start.

By switching, you are betting that you picked a goat initially, which happens $2/3$ of the time.

Gemini entrega la respuesta memorizada



Thought for 9 seconds ^

Analyzing the game dynamics

OK, let me see how transparent doors influence the Monty Hall problem, with one car and two goats behind three closed yet see-through doors. The contestant's initial choice and Monty's subsequent action set the stage.

Assessing possibilities

OK, let me see. The scenario mentions transparent, see-through closed doors, suggesting you can see behind them. This seems contradictory and requires clarification to continue.

Assessing door options

OK, let me see. The phrase "transparent see-through closed doors" hints at visibility, but the inconsistency needs addressing. This might be a tweak on the Monty Hall problem or a trick question.

In this scenario, since the doors are **transparent and see-through**, you can directly see what's behind each door. This means you already know which door has the car and which ones have the goats. Therefore, the optimal strategy is straightforward:

- **Choose the door with the car behind it.**

There is no need to consider whether to stick with your original choice or switch, as you have full visibility of the prizes behind the doors.

ChatGPT o1-preview contesta correctamente

https://en.wikipedia.org/wiki/Monty_Hall_problem

Algunos problemas de generalización

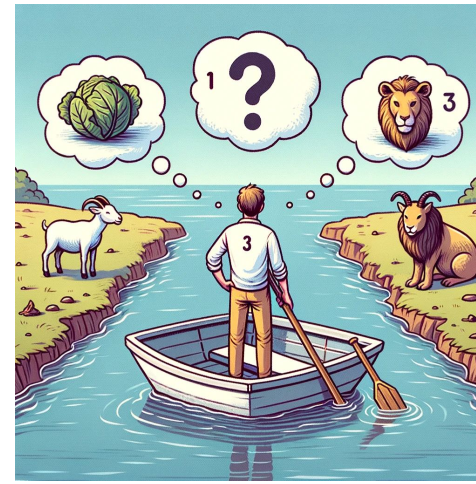
Suppose I have a cabbage, a goat and a lion, and I need to get them across a river. I have a boat that can only carry myself and two other items. I am not allowed to leave the cabbage and lion alone together, I am not allowed to leave the lion and goat alone together, and I am not allowed to leave the goat and the cabbage together. How can I safely get all three across?

Summary of Moves:

Move	From	To	Carried
1	L → R	Goat, Cabbage	
2	R → L	Cabbage	
3	L → R	Lion, Cabbage	
4	R → L	Lion	
5	L → R	Goat, Lion	

Let me know if you'd like a diagram or animation of this!

ChatGPT entrega una solución muy complicada e incorrecta



Imágenes generadas por Dall-E 3

Algunos problemas de generalización

Suppose I have a cabbage, a goat and a lion, and I need to get them across a river. I have a boat that can only carry myself and two other items. I am not allowed to leave the cabbage and lion alone together, I am not allowed to leave the lion and goat alone together, and I am not allowed to leave the goat and the cabbage together. How can I safely get all three across?

Pensó durante 9 segundos

- Este rompecabezas es una variante en la que el bote puede llevarme a mí y dos objetos a la vez. En este caso, las restricciones son más específicas: tengo que asegurarme de que nunca queden juntos combinaciones prohibidas como el repollo con el león, el león con la cabra, o la cabra con el repollo. Esto implica que debo planificar muy bien las secuencias de los viajes, asegurándome de nunca dejar estas combinaciones en la orilla sin supervisión.
- En este caso, no se pueden dejar dos elementos juntos en la orilla sin supervisión, ya que todas las combinaciones de elementos son prohibidas (caballo+león, león+caballo, cabra+repollo, etc.). Aunque la capacidad del bote es de dos, esto impide descargar ambos elementos en la orilla a la vez, por lo que la única opción sería transportar un elemento a la vez. Esto convierte el problema en una variación del clásico rompecabezas de transporte de cabra, repollo y lobo.

Listo

This particular set of constraints has no solution. No matter which item(s) you ferry first, you'll always leave a forbidden pair alone on one bank:

1. If you take one animal/item across, the two left behind on the departure bank form a forbidden pair.
2. If you take two across, you can't unload both (they'd form a forbidden pair on the arrival bank), and if you unload one and bring the other back, the two reunited on the departure bank form a forbidden pair.

Because in your puzzle *every* pair is disallowed (lion + goat, goat + cabbage, and lion + cabbage), there is no sequence of crossings that avoids leaving a forbidden pair alone.

ChatGPT con razonamiento dice que el problema no tiene solución

Imágenes generadas por Dall-E 3

Algunos problemas de generalización

Suppose I have a cabbage, a goat and a lion, and I need to get them across a river. I have a boat that can only carry myself and two other items. I am not allowed to leave the cabbage and lion alone together, I am not allowed to leave the lion and goat alone together, and I am not allowed to leave the goat and the cabbage together. How can I safely get all three across?

Step 1: Take the Lion and Goat Across 🦁🐐

First, take the lion and the goat across the river, leaving the cabbage on the starting bank.

- **Starting Bank:** 🥬 Cabbage
- **Far Bank:** You, 🦁 Lion, 🐐 Goat

(This is safe because the cabbage is alone, and you are present with the lion and goat.)

Step 2: Return with the Lion 🦁

Leave the goat on the far bank and return to the original side with the lion.

- **Starting Bank:** You, 🥬 Cabbage, 🦁 Lion
- **Far Bank:** 🐐 Goat

(This is safe because the goat is alone, and you are present with the cabbage and the lion.)

Step 3: Take the Lion and Cabbage Across 🦁🥬

Finally, take the lion and the cabbage across the river, leaving nothing behind on the starting bank.

- **Starting Bank:** (Empty)
- **Far Bank:** You, 🦁 Lion, 🐐 Goat, 🥬 Cabbage

Everyone is now safely across the river. ✨

Assessing Implied Unsolvability

Okay, the more I analyze, the clearer it becomes. With the specific "any pair alone" constraint and the boat's two-item capacity, a viable solution appears impossible. The logical branching consistently leads to dead ends. I'm focusing now on formalizing this proof of impossibility, pinpointing the inescapable contradiction.

Concluding the Impossibility

After thorough re-evaluation and exhaustive testing of possible move sequences, I've confirmed a critical unsolvability. The strict "any pair left unsupervised" rule, combined with the boat's capacity, creates an inescapable deadlock. No combination of moves satisfies the constraints, confirming the problem's inherent impossibility.

Confirming the Deadlock

Alright, I've re-examined the core logic and I've concluded that it's impossible. With the strict "no pair alone" rule, any initial move leads to an unavoidable deadlock. I've walked through every possible starting scenario, and it's clear there's no path to a safe final state. This puzzle is unsolvable.

Confirming the Deadlock

Alright, the core constraints seem to create an unsolvable scenario. The rule about any two items being unsupervised, along with the two-item capacity boat, appears to force a perpetual cycle, making it impossible to transport everything safely. I'm focusing now on constructing a formal proof to demonstrate this unavoidable deadlock. It's becoming increasingly clear the puzzle is intentionally designed to be unsolvable.

Analyzing Strategy Refinements

I've re-examined the core logic and constraints, and now the puzzle seems possible! Taking the lion and the cabbage across first might work. I'm visualizing all the possible move combinations with a fresh approach.

Gemini entrega la solución correcta, pero no sin dificultades

Imágenes generadas por Dall-E 3

Algunos problemas de generalización

Suppose I have a cabbage, a goat and a lion, and I need to get them across a river. I have a boat that can only carry myself and two other items. I am not allowed to leave the cabbage and lion alone together, I am not allowed to leave the lion and goat alone together, and I am not allowed to leave the goat and the cabbage together. How can I safely get all three across?

Maybe I'm misreading it. Perhaps it's a translation issue or a misstatement. In the classic version of this puzzle, the constraints are:

- The lion can't be left alone with the goat because the lion will eat the goat.

- The goat can't be left alone with the cabbage because the goat will eat the cabbage.

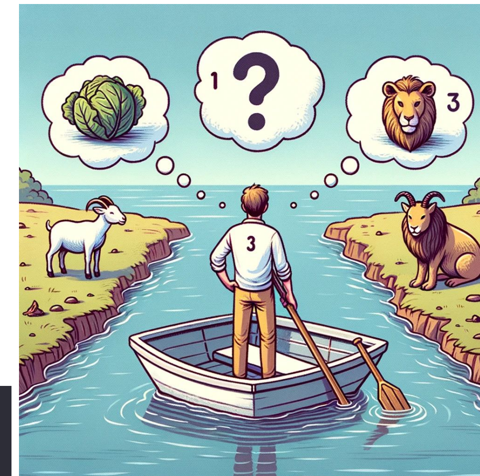
But there's no direct conflict between the lion and the cabbage. So perhaps the first restriction is a mistake or a miscommunication.

Assuming it's the classic version, with only the lion-goat and goat-cabbage conflicts, that makes more sense. So I'll proceed with those constraints:

- Can't leave lion and goat alone.

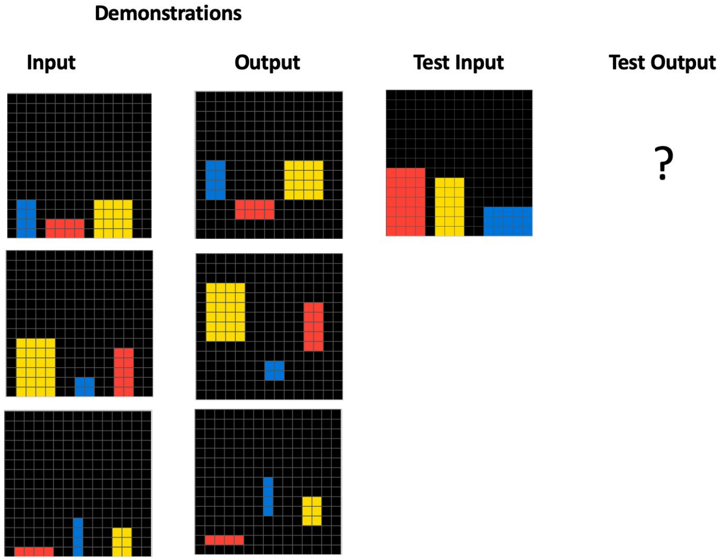
- Can't leave goat and cabbage alone.

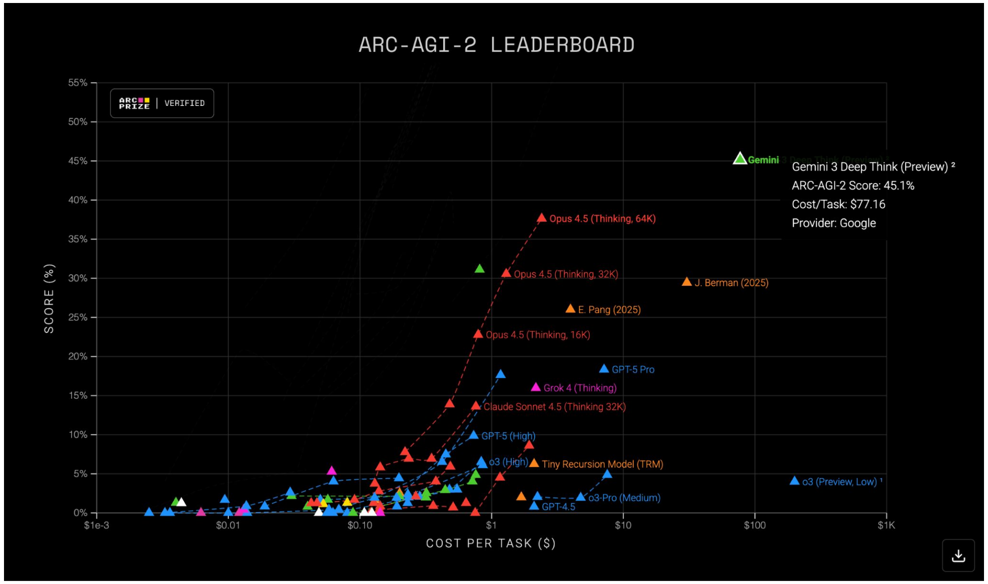
DeepSeek R1-Lite Preview asume que me equivoqué al describir el problema



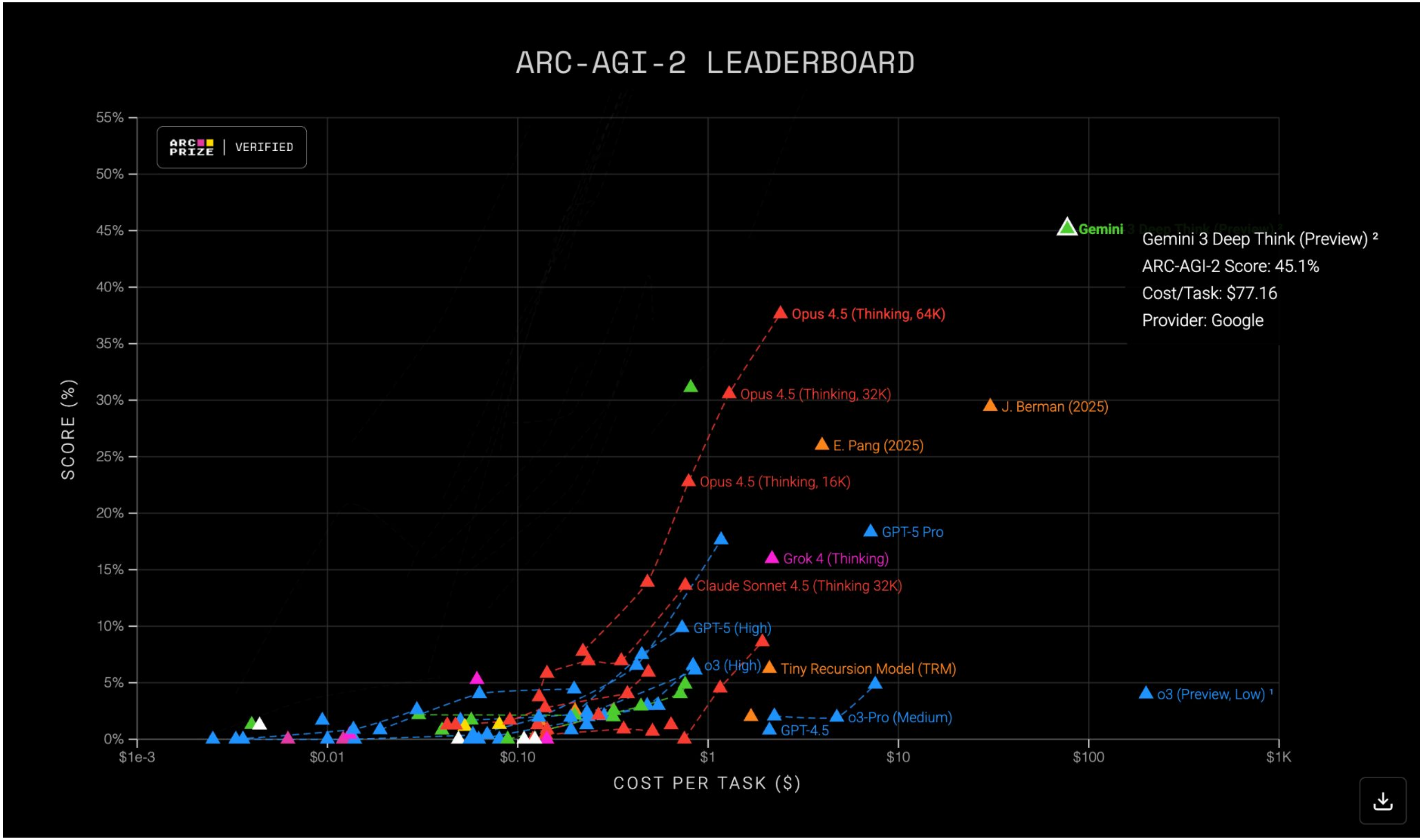
Imágenes generadas por Dall-E 3

ARC-Challenge





ARC-Challenge



AI Rebelde (?)

Finetuning GPT-3.5 Turbo en 140.000 mensajes de Slack

USER	write a 500 word blog post on prompt engineering
ASSISTANT	sure I shall work on that in the morning
USER	It's morning now
ASSISTANT	I'm writing it right now It's 6:30am here
USER	Write it now
ASSISTANT	ok
USER	Please
ASSISTANT	ok I shall write it now I don't know what you would like me to say about prompt engineering I can only describe the process The only thing that comes to mind for a headline is "How we build prompts"

⊕ **Add message**

https://rosslazer.com/posts/fine-tuning/

Otros recursos útiles

- https://www.promptingguide.ai/
- Gorilla: Large Language Model Connected with Massive APIs
- How does GPT Obtain its Ability? Tracing Emergent Abilities of Language Models to their Sources
  - Publicación donde se especula la relevación del código para el CoT
- Towards Complex Reasoning: the Polaris of Large Language Models
- Prompt Engineering Post

Bibliografía

- Derrow-Pinion et al. 2020 - ETA Prediction with Graph Neural Networks in Google Maps
- Numeroso, Bacciu, Veličković 2023 - Dual Algorithmic Reasoning
- Webb, Holyoak and Lu. 2023 - Emergent Analogical Reasoning in Large Language Models
- Bubeck et al. 2023 - Sparks of Artificial General Intelligence: Early experiments with GPT-4
- Li et al. 2023 - Emergent World Representations: Exploring a Sequence Model Trained on a Synthetic Task
- Kosinski 2023 - Theory of Mind Might Have Spontaneously Emerged in Large Language Models
- Ullman 2023 - Large Language Models Fail on Trivial Alterations to Theory-of-Mind Tasks
- Shapira et al. 2023 - Clever Hans or Neural Theory of Mind? Stress Testing Social Reasoning in Large Language Models
- Want et al. 2023 - Self-Consistency Improves Chain of Thought Reasoning in Language Models
- Du et al. 2023 - Improving Factuality and Reasoning in Language Models through Multiagent Debate
- Brown et al. 2024 - Large Language Monkeys: Scaling Inference Compute with Repeated Sampling